

FORM FOUR PHYSICS MARKING SCHEME MARCH 2025

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
D	E	B	D	B	E	B	E	C	A

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List A	i	ii	iii	iv	v	vi
List B	G	E	F	A	C	H

3 (a) Answer:

Angle between mirrors = 70°

Since the incident ray is parallel to A, it hits B at 70°

The law of reflection: angle of incidence = angle of reflection

So, the angle of reflection on mirror A = 70°

Answer: 70°

(b) Answer:

Let v = image distance

$$m = v/u = 7$$

$$\text{Total distance} = v + u = 160$$

$$v = 7u \rightarrow 7u + u = 160 \rightarrow 8u = 160 \rightarrow u = 20 \text{ cm} \rightarrow v = 140 \text{ cm}$$

Using lens formula:

$$1/f = 1/v + 1/u = 1/140 + 1/20 = (1 + 7)/140 = 8/140 = 1/17.5$$

$$f = 17.5 \text{ cm}$$

Answer: 17.5 cm

4 (a) Answer:

$$\text{Pressure difference} = 76 - 65 = 11 \text{ cm Hg} = 0.11 \text{ m}$$

$$\text{Using } h = P / (\rho g)$$

$$\rho = 13600 \text{ kg/m}^3, g = 9.81 \text{ m/s}^2$$

$$\text{Pressure} = 0.11 \times 13600 \times 9.81 = 14630.16 \text{ Pa}$$

$$\text{Height } h = P / (\rho g) = 14630.16 / (1.225 \times 9.81) = 14630.16 / 12.015$$

$$h \approx 1218 \text{ m}$$

Answer: Approximately 1218 m

(b) Answer:

Let x = force by Q, then force by P = $800 - x$

Taking moments about P's position:

$$x \times (2 + 3) = 800 \times 3 \rightarrow x \times 5 = 2400 \rightarrow x = 480 \text{ N}$$

$$P = 800 - 480 = 320 \text{ N}$$

Q supports 480 N, P supports 320 N

5(a) The recoil velocity of a gun is much less than velocity of bullet simply because, the gun has greater mass than bullet, since the linear momentum is directly proportional to the mass and its velocity.

(b) Data Given

Diameter of larger cylinder(D) =8x

Diameter of small cylinder(d) =x

Then, V.R = $D^2/d^2 = (8x)^2/x^2$

V.R = 64.

Efficiency =80 %

V.R =64

But ,

efficiency = MA/VR x 100%

90% = MA/64 x 100

M.A = 57.6

6.(a) The black color being a better absorber of heat, hence bulb Q gets more heated faster than P due to heat radiated by the heater, this resulting in the expansion of air inside bulb Q. The expanded air requires more space and thus pushes water towards X. Hence, the level of water in Y falls while that in X rises.

(b) The newly constructed Tanzania Standard gauge rails expand during days, hence rails gap become smaller due to expansion, while standard gauge rails contracts during winter days, hence rails gap become wider due to contraction.

7(a) Answer:

For a pipe closed at one end:

- Fundamental mode: 1st harmonic has $\frac{1}{4}$ wavelength in pipe

- First overtone: 3rd harmonic with $\frac{3}{4}$ wavelength in pipe

First overtone (3rd harmonic):

Answer:

Fundamental: $\frac{1}{4} \lambda$

First overtone: $\frac{3}{4} \lambda$

7 (b) Object appear coloured when light falls onto it because coloured objects tend to reflect light of its own colour falling onto it and absorb the rest

8. (a) The air in the refrigerator cabinet in contact with the freezer at the top is colder than the air below. It is therefore denser and sinks to the bottom. The less cold air rises to the top and becomes colder hence sinks. Therefore, the air in the refrigerator cabinet is thus kept cold with help of the freezer **(04 marks)**

(b) Total force stretching spring (F) = load + pan weight **(00 marks)** 2 1

$$= 2\text{N} + 0.4\text{N}$$

$$= 2.4\text{N} \text{ (00 marks)} \text{ 2 1}$$

From: Hookes' law (**where:** k is force constant) **(00 marks)** eF k = 2 1

$$\text{(00 marks)} \text{ mm N k } 244 .2 = 2 \text{ 1}$$

$$= 0.1 \text{ N/mm (00 marks) 2 1}$$

Given: e = 16mm to find F

From: eF k =

$$\text{(00 marks)} \text{ mm mmN ke F } 16/1.0 = 2 \text{ 1}$$

$$= 1.6\text{N} \text{ (00 marks) 2 1}$$

Load = F- pan weight **(00 marks)** 2 1

$$= 1.6\text{N} - 0.4\text{N}$$

$$= 1.2\text{N} \text{ (00 marks) 2 1}$$

The load on the scale pan is 1.2N

9. (a) Answer:

Charging Circuit Diagram:

E

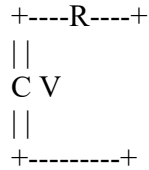
(+)-S-R-+-C-(-)

||

+---V---+

During charging, when the switch is closed, current flows and the capacitor stores charge. The voltage across the capacitor increases gradually following an exponential curve, approaching the battery voltage. The current decreases as the capacitor becomes fully charged.

Discharging Circuit Diagram:



During discharging, the capacitor is disconnected from the cell and allowed to release stored energy through the resistor. The voltage across the capacitor decreases exponentially over time, and the current also decreases.

(b) Answer:

Since the wires are of same material and length, resistance is inversely proportional to cross-sectional area.

If $A_1 : A_2 = 2 : 1$, then $R_1 : R_2 = 1 : 2$

Using Ohm's Law, $I = V/R$

So current is inversely proportional to resistance:

$I_1 : I_2 = R_2 : R_1 = 2 : 1$

Thus, wire A carries twice the current of wire B.

(c) Answer:

The voltage across the load is $V = E - Ir$, where E is the EMF, I is current, and r is internal resistance.

If internal resistance r is high, more voltage is lost inside the source and less is available for the external circuit. A low internal resistance ensures maximum voltage is delivered to the load.

10 . (a) 10. (a) When hall has many people, sound is absorbed by clothes and skin of audience thus echoes do not

occur **but** when the hall has few people the sound is less absorbed and multiple echoes arises which hinders clear audibility of sound. **(03 marks)**

b) Consider a string fixed at point A and B

Fundamental frequency () $0 f$

Length (L) (00 marks) 2) (□ wavelength= 2 1

Velocity = wavelength frequency (□) f

but: □ $V f = 0 L 2 =$ □

11 . (a) Answer:

For a tube closed at one end, the resonance occurs at odd multiples of quarter wavelengths.

Shortest length L is given by:

$$L = \lambda / 4$$

From $L = 160 \text{ mm} = 0.160 \text{ m}$ for $f = 500 \text{ Hz}$

Then $\lambda = 4 \times 0.160 = 0.64 \text{ m}$

Speed of sound, $v = f \times \lambda = 500 \times 0.64 = 320 \text{ m/s}$

Now for $f = 800 \text{ Hz}$:

$$\lambda = v / f = 320 / 800 = 0.4 \text{ m}$$

Shortest length $= \lambda / 4 = 0.4 / 4 = 0.1 \text{ m} = 100 \text{ mm}$

Answer: 100 mm

(b) Answer:

Lenz's law states that the direction of induced current is such that it opposes the change that caused it.

This opposition ensures that external work is needed to produce the current, meaning energy is not

created but converted—typically from mechanical to electrical energy. Thus, it conserves energy by preventing spontaneous energy gain, which would violate the law of conservation of energy.

(c) (i) Answer:

When the switch is closed, current flows through the coil, magnetizing the core and attracting the armature A. This causes the striker to hit the gong. As A moves, it breaks the circuit, demagnetizing the core, and the armature returns due to the spring, repeating the process rapidly to produce a ringing sound.

(ii) Answer:

The steel strip S acts as a spring to pull the armature A back to its original position after it has been attracted by the electromagnet. It also helps break the circuit, stopping the current, so the cycle can repeat and the bell rings continuously as long as the switch remains closed.